

Statistics Quick Reference Guide

A companion guide to the Statistics Interview Question Bank — read this first, then use the question bank to practice applying these ideas to business scenarios.

1. Descriptive Statistics

Descriptive statistics summarize a dataset in a few numbers so you can understand it quickly, without looking at every single value.

Measures of Central Tendency

These answers: "What's a typical value in this data?"

- **Mean (average):** Add up all values, divide by the count. Sensitive to extreme values — one very high salary can pull the mean up a lot.
- **Median:** The middle value when data is sorted. Not affected by extreme values — this is why "median household income" is reported more often than "average household income."
- **Mode:** The most frequently occurring value. Most useful for categorical data (e.g., most common product size ordered).

When to use which: If the data has outliers or is skewed (like salaries, house prices, or spending), the median usually represents "typical" better than the mean. If the data is roughly symmetric with no extreme values, the mean works well and uses more information.

Measures of Spread (Dispersion)

These answers: "How much do values differ from each other?"

- **Range:** Maximum minus minimum. Simple but very sensitive to a single extreme value.
- **Variance:** The average of squared differences from the mean. Tells you how spread out the data is, but the units are squared (e.g., "rupees squared"), which makes it hard to interpret directly.
- **Standard Deviation (SD):** The square root of variance — brings the units back to the original scale (e.g., rupees). A low SD means values cluster tightly around the mean; a high SD means values are spread widely.

Why this matters in business: Two datasets can have the same mean but very different spread. A factory where all rods are within 1mm of the target is very different from one where the average is the same but individual rods vary by 10mm — even though the *mean* looks identical, the *spread* tells the real story.

Distribution Shape

- **Normal distribution:** Symmetric, bell-shaped — most values cluster near the mean, with fewer values as you move further away.
- **Skewness:** When data leans to one side.
 - **Right-skewed (positive skew):** A long tail of high values — e.g., income or delivery times (most deliveries are quick, but a few take much longer due to traffic).
 - **Left-skewed (negative skew):** A long tail of low values — less common in business data, but can appear with things like exam scores where most students do well and a few score very low.
- **Outliers:** Values that are unusually far from the rest of the data. They can be genuine (a huge one-time purchase) or errors (a data entry mistake). Always investigate before deciding whether to keep, remove, or treat them separately.

Visualizing Distributions

- **Histogram:** Shows the shape of the data — where values cluster, and whether it's skewed.
- **Box plot:** Shows median, spread (quartiles), and outliers in one compact picture — very useful for comparing groups (e.g., waiting times across departments).

2. Probability Basics

Probability measures how likely something is to happen, on a scale from 0 (impossible) to 1 (certain).

Core Rules

- **Independent events:** The outcome of one event doesn't affect the other. Flipping a coin twice — the second flip doesn't "remember" the first.
- **Dependent events:** One event's outcome affects the probability of the other. Drawing cards from a deck without replacement is dependent, because the deck changes after each draw.
- **Conditional probability:** The probability of one event *given that* another has already happened. Written as $P(A | B)$ — "the probability of A, given B."

Common Business Traps

- **Gambler's fallacy:** Believing that past independent events affect future ones — e.g., assuming three failed transactions in a row means the fourth is "due" to succeed, when each transaction's success is actually independent of the last.
- **Base rate neglect:** Ignoring how rare or common something is overall. A spam filter that's "97% accurate" sounds great, but if spam is only 2% of all emails, that accuracy

number alone doesn't tell you how well it actually catches spam versus how often it wrongly flags real emails.

- **Correlation in behavior $\neq$ causation:** If people who watch sci-fi also tend to watch documentaries, that's a relationship between two behaviors, not proof that one causes the other.

3. Sampling Techniques

Since it's rarely possible or affordable to collect data from an entire population, we collect a **sample** — a smaller group — and use it to make conclusions about the whole.

Why Sampling Matters

A sample is only useful if it **represents** the population it's meant to describe. A large sample that's biased is often worse than a smaller, well-chosen one — size doesn't fix bias.

Common Sampling Methods

- **Random sampling:** Every member of the population has an equal chance of being selected. Reduces bias but requires a complete list of the population to draw from.
- **Stratified sampling:** The population is divided into subgroups (strata) — e.g., by age group or region — and samples are drawn from each subgroup proportionally. Useful when you want to make sure smaller subgroups aren't missed.
- **Cluster sampling:** The population is divided into clusters (e.g., cities), and a few whole clusters are randomly selected and fully surveyed. Useful when a population is spread across a large area and full random sampling is impractical.

Common Sources of Sampling Bias

- **Selection bias:** The way the sample was chosen systematically favors certain outcomes (e.g., surveying only customers who attended a workshop, then generalizing to all employees).
- **Self-selection bias:** People who choose to respond (like leaving a review) may differ systematically from those who don't — usually people with strong opinions, positive or negative, are more likely to respond.
- **Survivorship bias:** Only looking at things that "survived" a process, ignoring the ones that didn't — e.g., studying only successful companies and missing the failed ones that might reveal more useful patterns.

In practice: Whenever you see a statistic from a survey or study, a natural first question is: "Who exactly was sampled, and could that group be different from who we're trying to draw conclusions about?"

4. Correlation & Relationships

Correlation

Correlation measures the strength and direction of a relationship between two numeric variables, usually on a scale from -1 to +1.

- **Positive correlation:** As one variable increases, the other tends to increase too (e.g., advertising spend and sales).
- **Negative correlation:** As one variable increases, the other tends to decrease (e.g., price and demand, in many cases).
- **No correlation:** No consistent relationship between the two.

Correlation vs Causation

This is one of the most important ideas in statistics: **just because two things move together doesn't mean one causes the other**. Classic example: ice cream sales and umbrella sales both rise in the rainy season — but ice cream doesn't cause rain, and umbrellas don't cause ice cream sales. A third factor (the season) drives both.

Before claiming causation, consider:

- Could a third, hidden factor be driving both variables?
- Does the timing make sense (does the proposed cause happen *before* the effect)?
- Has this relationship been tested with a controlled experiment, or only observed in existing data?

Covariance

Covariance also measures how two variables move together, but unlike correlation, it isn't standardized — its value depends on the units of the variables, which makes it hard to compare across different datasets. Correlation solves this by standardizing covariance to a fixed -1 to +1 scale, which is why correlation is used far more often in practice.

5. Quick Reference: A Few More Terms to Know (For Awareness)

The sections above are the core of what this course covers. A few interviewers may still casually bring up the terms below. You don't need to calculate anything here — just be able to explain what each means in plain language.

Statistical Significance When someone says a result is "statistically significant," they mean the difference observed (like a 4% sales increase) is unlikely to have happened just by chance — it's probably a real effect, not random noise. It does *not* automatically mean the difference is large or important for the business — a tiny, real difference can still be "significant."

Null Hypothesis vs Alternate Hypothesis Before testing anything, you start by assuming nothing has changed — that's the **null hypothesis** (e.g., "the new website design has no effect on conversion rate"). The **alternate hypothesis** is the claim you're actually trying to find evidence for (e.g., "the new design does change conversion rate"). The test tells you whether you have enough evidence to reject the null hypothesis.

P-value The p-value is a number between 0 and 1 that tells you how surprising your result would be *if* nothing had actually changed. A small p-value (commonly under 0.05) suggests the result is unlikely to be random chance, so you have some evidence against the null hypothesis. A large p-value means the result is quite plausible even if nothing changed — so you don't have strong evidence of a real effect.

Confidence Interval A confidence interval gives a range instead of a single number, along with how sure you are that the true value falls inside it. For example, "customer satisfaction is 82%, with a 95% confidence interval of 79%–85%" means: if you repeated this survey many times, about 95% of the intervals you'd calculate would contain the true satisfaction level. It's a way of expressing uncertainty honestly, instead of pretending a single sampled number is exact.

Central Limit Theorem (CLT) — one-line version Even if the underlying data is messy or skewed (like customer spending, which usually has a few very high spenders), the *average* of many random samples tends to follow a nice, predictable bell-shaped pattern as the sample size grows. This is why analysts can still make reliable estimates about averages even when the raw data itself isn't neatly distributed.